

# DAVID RAJ J *AI Engineer*

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 Chennai

 Github

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## PROFILE

**AI Engineer** with **3 years** of experience specializing in Generative AI, Retrieval-Augmented Generation (RAG), intelligent agent workflows, and multi-tool orchestration. Expert in building custom MCP (Model Context Protocol) servers and integrating them into production chatbots for secure interaction with Gmail, Google Drive, Google Calendar, Dropbox, and domain-specific APIs. Adept at engineering scalable RAG pipelines, ReAct-driven agent workflows, and webhook-based tool execution for real-world automation. Proven ability to deliver high-performance AI systems, achieving up to 38% higher retrieval precision, 42% lower latency, and 40% reduced analyst effort. Proficient in Transformers, Python, LangChain, LangGraph, and ML deployment on AWS.

## SKILLS

Python

SQL

MongoDB

EDA

Tensorflow

Pytorch

NLP

LLMs

MLops

Langchain

Langraph

Supervised and UnSupervised Learning

Flask

FastApi

RAG Workflow

LLM Fine tuning

Docker

MCP Tool Orchestration

ML-flow

AWS ECR-ECS-Sagemaker-Bedrock

Linux

## PROFESSIONAL EXPERIENCE

- Contus Tech, AI Engineer**

06/2025 – present | Chennai, India

  - Developed and deployed custom MCP servers enabling secure, tool-based interactions with Gmail, Google Calendar, Google Drive, Dropbox, internal enterprise APIs, and external STT/TTS voice pipeline servers.
  - Integrated MCP tools directly into LangGraph agent nodes, enabling dynamic context-aware tool calling, voice-agent communication, and multi-step automated workflows.
  - Built webhook-driven action tools using LLM function-calling to extract structured parameters and execute backend API workflows across enterprise systems.
  - Engineered a highly scalable FastAPI async backend supporting large volumes of concurrent LLM, MCP, and distributed voice-agent requests.
  - Designed an enterprise-grade RAG workflow delivering citation-backed responses with 38% higher retrieval precision.
  - Optimized vector DB performance (Chroma, Pinecone), reducing latency by 42% and doubling throughput.
- WellSpring Systems Pvt Ltd, ML Engineer**

03/2023 – 05/2025 | Chennai, India

  - Built **autonomous GenAI agents** for phishing and XSS detection using **LSTM neural networks**, achieving **89% accuracy** and reducing false positives by **20%**.
  - Developed a **LangGraph-powered RAG system** integrating cybersecurity playbooks to improve analyst response time and ensure consistent incident handling across **10+ scenarios**.
  - Created **LLM-powered summarization and analytics tools** enriched with MITRE ATT&CK, CVEs, and tactics, reducing manual reporting effort by **40%**.
  - Implemented a **threat reasoning engine** leveraging LLMs for automated alert correlation and suggested actions, reducing MTTI by **35%**.

## PROJECTS

- Study Mate AI**

01/2026 – 02/2026

**Skills:** LLMs, RAG, LangChain, FastAPI, React, MongoDB, ChromaDB, Docker

  - Built an **AI-powered learning backend** using **FastAPI and LangChain**, enabling AI tutoring, concept explanation, quiz generation, and PDF-based question answering powered by large language models.
  - Implemented a **Retrieval-Augmented Generation (RAG) pipeline** using **ChromaDB vector database and embedding models** to perform semantic search over educational documents and improve contextual answer generation.
  - Containerized the entire backend system using Docker**, enabling reproducible deployments, scalable microservice integration, and seamless interaction with vector databases and external AI model providers.
- MCP-Powered Multi-Tool Agent**

11/2025 – 12/2025

**Skills:** MCP, LangChain, LangGraph, Tool Orchestration

  - Created a unified AI workspace enabling a single conversational agent to access Gmail, Google Calendar, Google Drive, and Dropbox through **custom MCP servers**.
  - Implemented **context-aware tool routing** based on user intent and workflow state.
  - Added robust message parsing and input validation for safe parameter extraction before tool execution.
  - Leveraged **LangGraph** to maintain workflow consistency, tool call traceability, and error recovery pathways.
- KnowYourBite**

03/2025 – 04/2025

**Skills:** PaddleOCR, Python, FastAPI, Streamlit, OpenCV, Docker, AWS ECR, AWS ECS, Web Scraping, NLP

  - Designed and developed an AI-based application using **PaddleOCR** to extract nutritional and ingredient information from food packaging images, including automated detection of tabular data from product labels (e.g., chips packets) with improved text recognition for complex layouts.
  - Integrated real-time web search and data aggregation to analyze ingredient health risks, consumption limits, and age-based dietary recommendations, and built an interactive **Streamlit** frontend for image upload, analysis, and visualization of health insights.
  - Containerized the application using **Docker Compose** and deployed scalable services on **AWS ECS** with images stored in **AWS ECR**, enabling public access via cloud-hosted endpoints and reliable end-to-end processing.

EDUCATION

|                                                                                  |   |                                    |
|----------------------------------------------------------------------------------|---|------------------------------------|
| Scaler Academy, Advanced Data Science and Machine Learning                       | . | 11/2023 – 06/2025   Chennai, India |
| Guvi Geek Network, IIT-M Advanced Programming Professional & Master Data Science |   | 06/2021 – 11/2022   Chennai, India |
| Saveetha Engineering College, Bachelor of Engineering in EEE                     |   | 08/2012 – 04/2016   chennai, India |

LANGUAGES

|         |           |       |           |
|---------|-----------|-------|-----------|
| English | ● ● ● ● ● | Tamil | ● ● ● ● ● |
|---------|-----------|-------|-----------|